LAB EXPERIMENT

ITA0506-COMPUTER VISION

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EXPNO:3

AIM: Perform basic Image Handling and processing operations on the image

- Read an image in python and Convert an Image to show outline using Canny function.

Algorithm:

- import library
  Start Start the program and import the OpenCV module using import cv2.
- Read the input image
  Read the image from the given file path using cv2.imread().
- Check image loading
  Verify that the image is loaded (not None); if loading fails, print an error message and stop the program.
- Convert to grayscale
  Convert the color image to a grayscale image using cv2.cvtColor(img, cv2.COLOR_BGR2GRAY).
- Filter noise and detect edges
  Apply Gaussian blur on the grayscale image using cv2.GaussianBlur() and then perform Canny edge detection using cv2.Canny() to obtain the outline.
- Display images and exit
  Display the original and edge-detected images using cv2.imshow(), wait for a key press with cv2.waitKey(0), then close all windows using cv2.destroyAllWindows().

Code:

ita0506 lab 3.py - C:/Users/POORNA SKHT/OneDrive/Desktop/POORNA SKHT/ita0506 lab 3.py (3.13.5)

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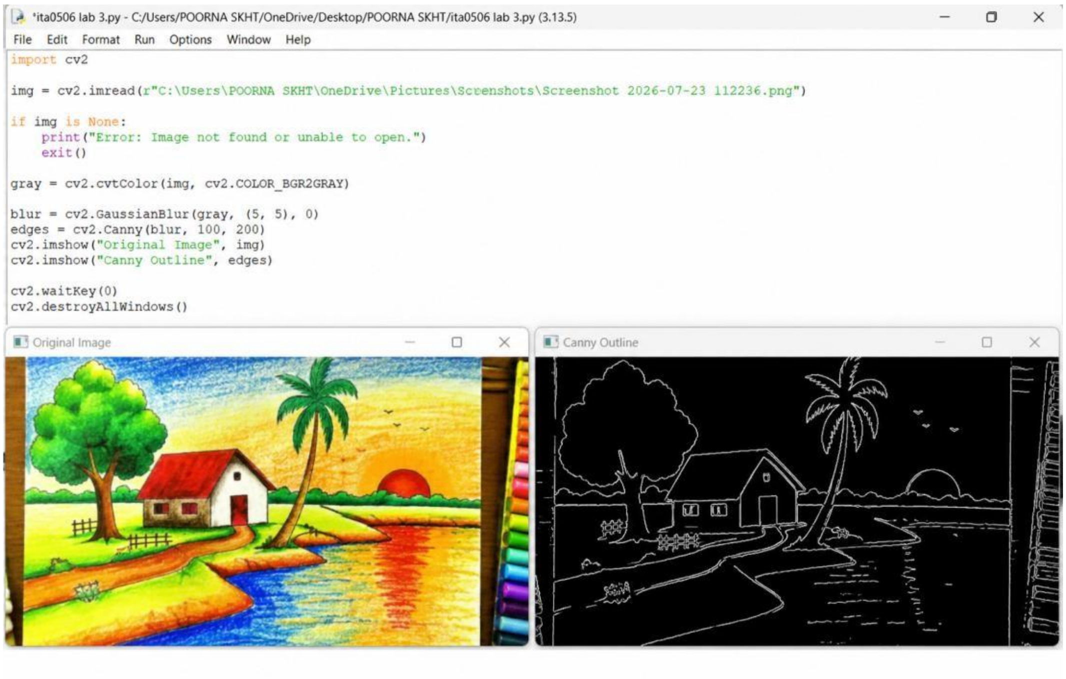
```python
import cv2

img = cv2.imread(r"C:\Users\POORNA SKHT\OneDrive\Pictures\Screenshots  Screenshot 2026-07-23 112236.png")

if img is None:
    print("Error: Image not found or unable to open.")
    exit()

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

blur = cv2.GaussianBlur(gray, (5, 5), 0)
edges = cv2.Canny(blur, 100, 200)
cv2.imshow("Original Image", img)
cv2.imshow("Canny Outline", edges)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

OUTPUT :



Result:

Thus, the input image was successfully read and processed using the **Canny Edge Detection** function in Python with OpenCV. The Canny algorithm detected the edges of the objects in the image and displayed the outline of the image successfully.